WEEK 7 - HEMATOLOGY • LECTURE 5 • SLIDE COMPANION

Hemolytic Anemias

Mark Dalgetty • designed to follow alongside the 51-slide lecture deck

Official Week 7 LOs | Slide-order companion | 5 ScholarRx Bricks
Emergency-diagnosis emphasis | Faculty companion gap-fill | Factoid sheet companion

Overview: Official Week 7 Learning Objectives

LO 1. Define hemolytic anemia and recognize its pathognomonic features.

LO 2. Identify the causes of hemolytic anemia and list its different classifications.

LO 3. Work up a suspected hemolytic anemia case.

LO 4. Recognize emergency hemolytic anemia diagnoses.

LO 5. Discuss treatment of select types of inherited hemolytic anemias.

LO 6. Discuss treatment of select types of acquired hemolytic anemias.

LO 7. Distinguish the different types of hemolytic transfusion reactions.

SCHOLARRX BRICKS PAIRED WITH THIS LECTURE

Red Blood Cell Metabolism & Hemolytic Anemia
Hereditary Spherocytosis
Microangiopathic Hemolytic Anemia
Autoimmune Hemolytic Anemias
Blood Transfusion

LO 1-2

Slides 3-12: normal RBC aging, hemolysis labs, intravascular vs extravascular, acute/chronic, hereditary/ acquired, intrinsic/extrinsic, immune/ nonimmune.

LO 3

Slides 13-20: history, exam, smear, emergency screen, DAT/Coombs.

LO 5

Slides 21-32: hereditary spherocytosis and G6PD deficiency.

LO 6

Slides 33-39: warm AIHA vs cold agglutinin disease.

LO 4

Slides 40-50: MAHA, schistocytes, TTP, PLASMIC score, empiric treatment.

LO 7

Official-LO gap-fill: acute vs delayed hemolytic transfusion reactions are developed in the uploaded companion handout rather than the slide deck.

SOURCE NOTE

The current 51-slide deck is Mark Dalgetty's lecture. The uploaded companion DOCX says it was based on a 2026 lecture by Caroline H. Hana. Because the official objectives line up closely, I use it only as a **supplemental explanation source**, especially for the transfusion-reaction objective that the slide deck itself does not develop. Slide-derived and companion-derived material are kept distinct.

THE WHOLE LECTURE IN ONE LINE

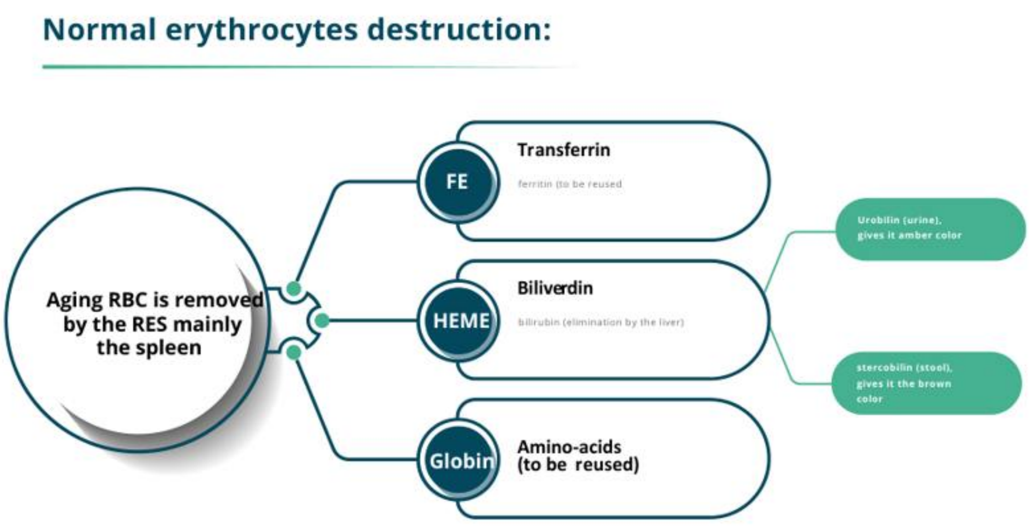
Prove hemolysis -> decide where/how it is happening -> rule out emergencies -> use smear + DAT + history to identify the mechanism.

Slides 3-6 | **LO 1**

Normal RBC aging and the laboratory signature of hemolysis

Slide 3: a normal RBC survives about 120 days

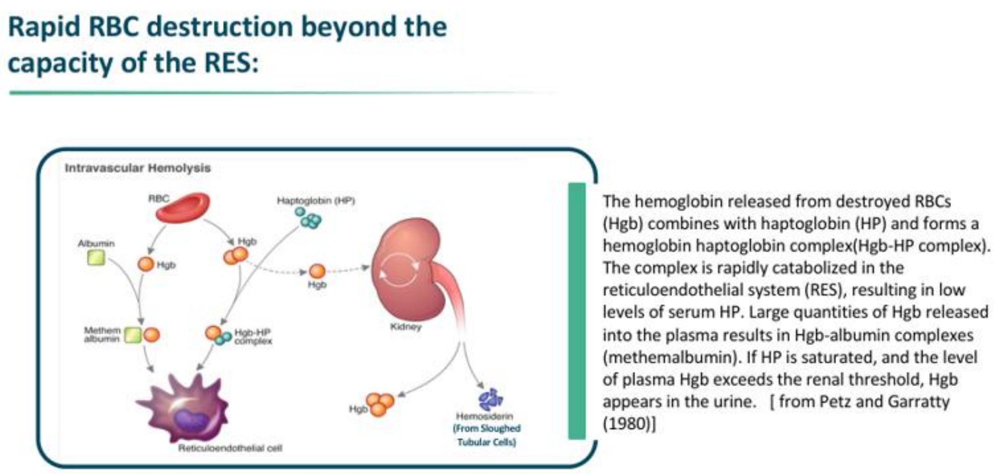
- RBC survival depends on **membrane flexibility**, functioning metabolic enzymes, and hemoglobin remaining soluble/reduced.
- With aging, enzymes decay, oxidized hemoglobin can precipitate as Heinz bodies, and membrane changes mark the cell for removal.
- Senescent RBCs are removed mainly by the **reticuloendothelial system, especially splenic macrophages**.



Lecture slide 4 - full slide retained. Normal extravascular clearance recycles iron through transferrin, globin into amino acids, and heme pigment into bilirubin products. This explains why hemolysis raises unconjugated bilirubin.

WHAT THE BILIRUBIN IS TELLING YOU

More RBC destruction means more heme breakdown. The liver may be unable to conjugate the increased bilirubin load fast enough, producing **indirect/unconjugated hyperbilirubinemia, jaundice, and chronic pigment-gallstone risk**.



Lecture slide 5 - full slide retained. When RBCs lyse in plasma, free hemoglobin binds haptoglobin. The complex is rapidly cleared, so serum haptoglobin falls; if binding capacity is exceeded, hemoglobin can appear in urine.

NEED TO MEMORIZE - CORE HEMOLYSIS LABS

Reticulocytes up + LDH up + indirect bilirubin up + haptoglobin down.
The lecture's definition also requires anemia / accelerated RBC destruction.

HAPTOGLOBIN NUANCE

Low haptoglobin supports hemolysis, but a **profoundly low/undetectable value is especially suggestive of intravascular hemolysis**, where large amounts of hemoglobin spill directly into plasma.

LO 1 checkpoint

Can you explain each classic hemolysis lab direction mechanistically rather than memorizing arrows?

Slides 7-12 Classify hemolysis five different ways LO 2

Classification	Side A	Side B	High-yield examples
Location	Extravascular	Intravascular	HS/warm AIHA vs MAHA/ mechanical/complement/toxin
Tempo	Acute	Chronic	G6PD/TTP/AHTR vs HS/ device-related chronic hemolysis
Origin	Hereditary	Acquired	HS/G6PD vs AIHA/TTP/device
RBC locus	Intrinsic / intracorpuscular	Extrinsic / extracorpuscular	Membrane/enzyme/Hb defect vs antibody/shear/infection
Immune?	Immune-mediated	Nonimmune	DAT-positive AIHA vs HS/ G6PD/MAHA

Slides 8-9: intravascular vs extravascular and acute vs chronic

- **Extravascular:** primarily spleen/liver macrophage clearance. Think poorly deformable cells or antibody-coated cells.
- **Intravascular:** membrane rupture inside vessels from mechanical shear, complement MAC, overwhelming injury, or toxins/infection.
- **Acute:** rapid Hb drop and symptoms are more dramatic.
- **Chronic:** more compensation, but ongoing bilirubin/gallstone/splenomegaly consequences may accumulate.

LOCATION SHORTCUT

Spleen eating cells = extravascular.
Cells bursting in vessels = intravascular.

BOARD ENRICHMENT - PNH RECOGNITION ONLY

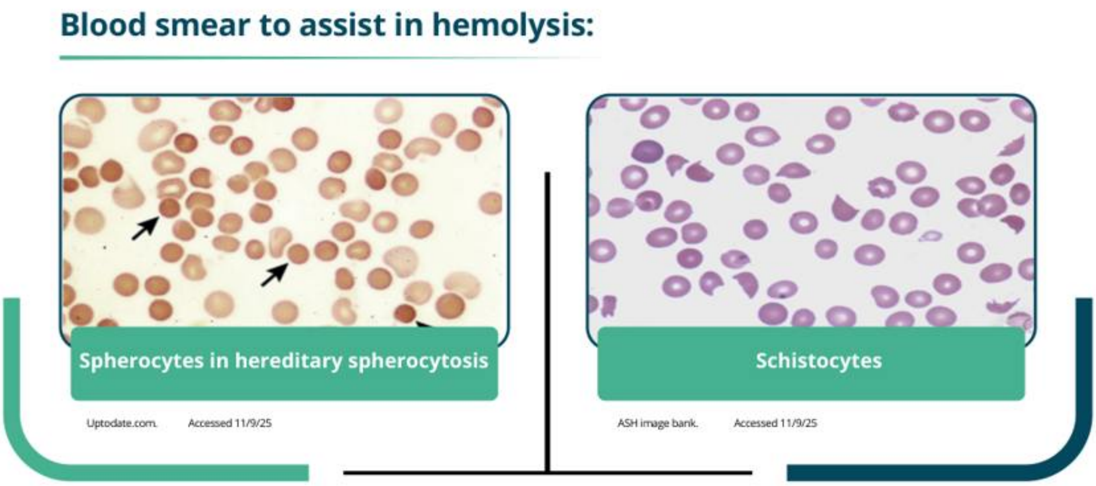
The lecture briefly places PNH among intrinsic membrane-related causes despite being acquired. Recognition-level board fact: loss of GPI-linked complement regulators **CD55/CD59** makes RBCs complement-sensitive and produces intravascular hemolysis. The deck does not develop PNH further, so it is not a major focus here.

Slides 13-20 | Work-up: history -> smear -> emergency screen -> DAT | LO 3 + LO 4

Slide 13: history can identify the mechanism before a specialized test

- Recent transfusion -> hemolytic transfusion reaction.
- Recent surgery, prosthetic valve, or intravascular device -> mechanical hemolysis.
- Recent drug/oxidant exposure -> G6PD or drug-induced immune hemolysis.
- Family history / childhood disease -> hereditary cause.
- Race/ancestry and onset can raise or lower pretest probability for inherited disorders.

Slide 15-17: the smear is a mechanism detector



Lecture slide 17 - full slide retained. Spherocytes suggest membrane loss/extravascular destruction (HS or warm AIHA); schistocytes indicate mechanical fragmentation such as MAHA/DIC. The DAT and clinical context distinguish causes.

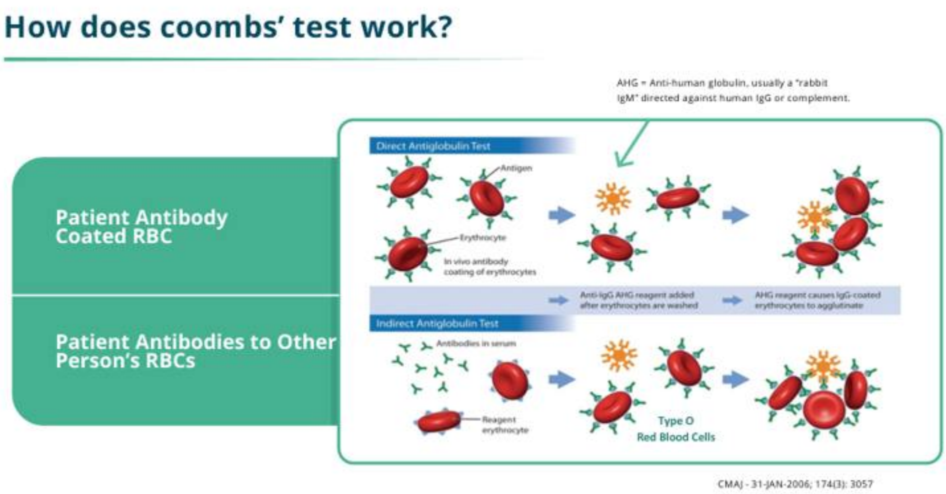
Smear finding	Think	Next discriminating clue
Spherocytes	HS or warm AIHA	**DAT negative** favors HS; **DAT positive** favors AIHA.
Schistocytes	MAHA / DIC / mechanical shear	Platelets, coagulation tests, renal/neuro findings, emergency evaluation.
Bite cells	G6PD deficiency	Oxidant trigger + Heinz-body biology.
Intracellular parasites	Malaria / babesiosis	Exposure/travel + species-specific testing.
Ghost cells	Severe intravascular lysis	Lecture specifically mentions clostridial sepsis as a clue.

FACULTY EMERGENCY SCREEN - SLIDE 18

Before leisurely sorting the differential, look for **thrombosis, acute renal failure, very low hemoglobin, thrombocytopenia, or a recent transfusion**. Dalgetty explicitly ties these to urgent entities such as TMA/TTP and hemolytic transfusion reaction.

Slide 19: once immediate emergencies are considered, use DAT

- **DAT / direct Coombs positive:** supports immune-mediated hemolysis.
- **DAT negative:** search for nonimmune causes such as membrane/enzyme defects, mechanical fragmentation, infection, etc.



Lecture slide 20 - full slide retained. Direct testing asks whether immunoglobulin/complement is already attached to the patient's RBCs. Indirect testing detects antibodies in serum against test/donor RBC antigens.

NEED TO MEMORIZE - COOMBS

Direct = antibody/complement ON the patient's RBC.
Indirect = antibody in the patient's serum that CAN bind RBC antigens.

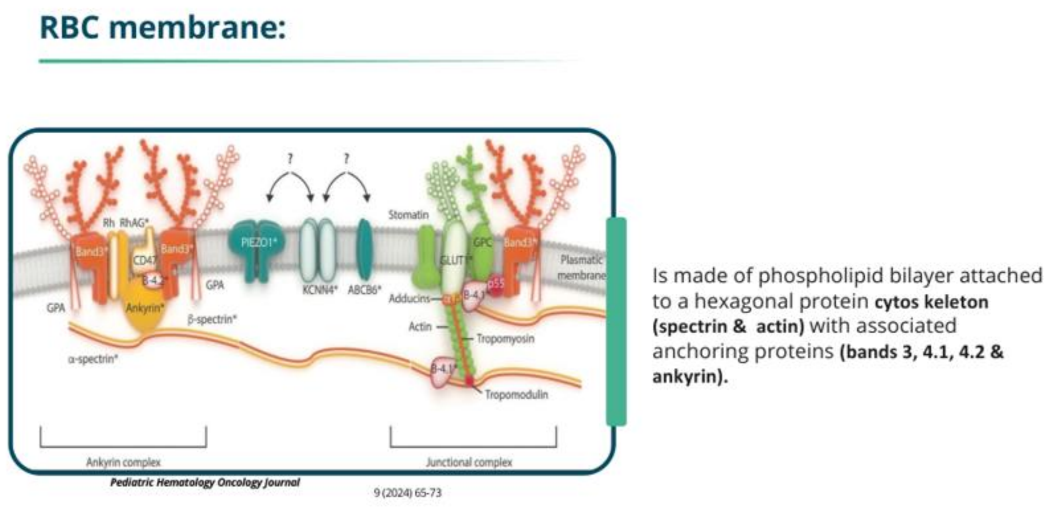
LO 3 checkpoint

Can you work a hemolysis vignette in order: confirm hemolysis -> inspect smear/ platelets/renal status -> ask about transfusion/device/drugs/family history -> DAT if immune disease is plausible?

Slides 21-32

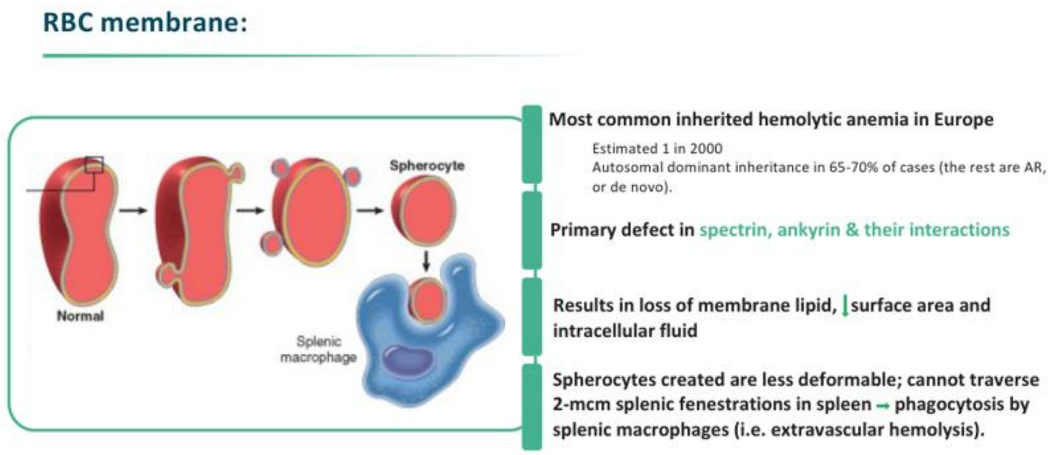
Slide 23 organizes inherited disease

Membrane: hereditary spherocytosis/elliptocytosis. **Metabolism:** G6PD/pyruvate kinase deficiency. **Hemoglobin:** HbS/HbC/unstable Hbs (mostly covered in the prior hemoglobinopathy lecture).



Lecture slide 24 - full slide retained. The membrane is a phospholipid bilayer anchored to a cytoskeletal network containing spectrin, actin, ankyrin, band 3 and associated proteins. Defects reduce deformability.

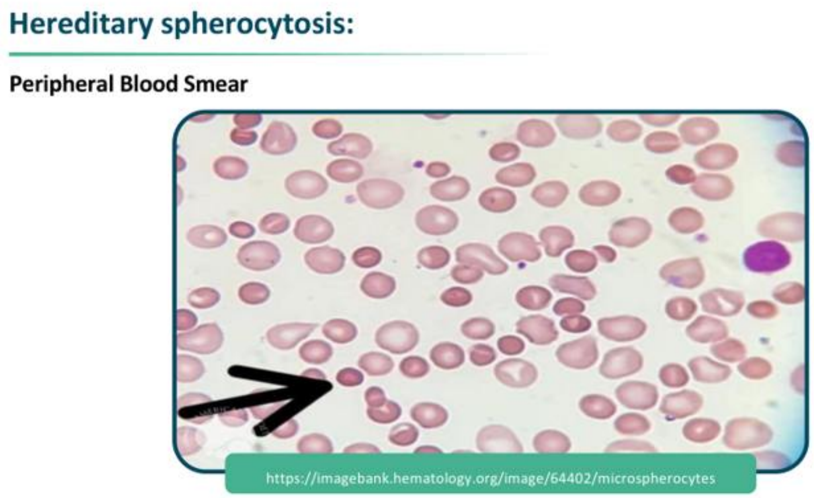
Hereditary spherocytosis



Lecture slide 25 - full slide retained. Loss of membrane surface area creates poorly deformable spherocytes that cannot traverse splenic fenestrations and are removed by splenic macrophages - classic extravascular hemolysis.

HS RECOGNITION SET

Spherocytes + high MCHC + splenomegaly/jaundice + family history + DAT negative.



Lecture slide 26 - full slide retained. Look for small dense round cells lacking normal central pallor. The morphology overlaps with warm AIHA, so the DAT is essential.

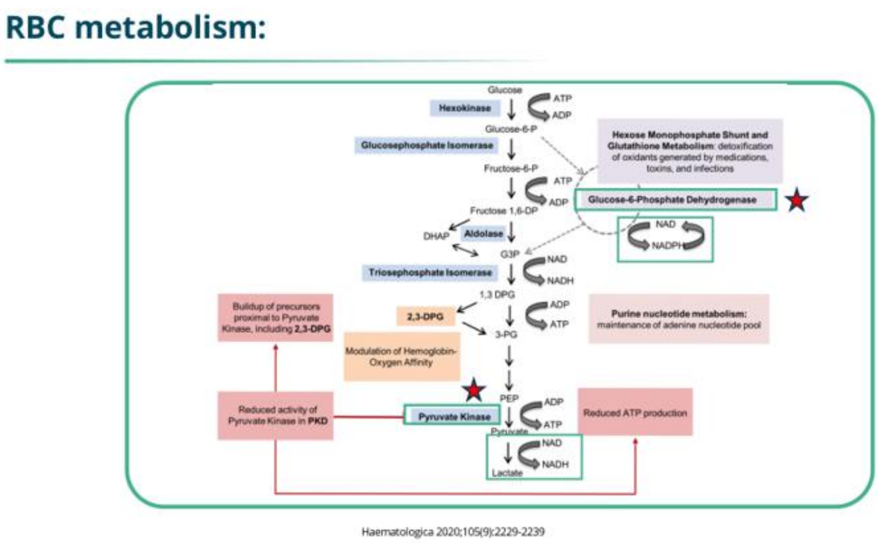
Slide 27-28: diagnosis and treatment

- The lecture teaches **increased osmotic fragility**: spherocytes lyse more easily in hypotonic saline.
- Supportive treatment includes folate and transfusion for severe anemia/aplastic crisis.
- Splenectomy can markedly reduce hemolysis; cholecystectomy may be indicated for pigment stones.
- Dalgetty advises deferring splenectomy in young children when possible.

CLINICAL/BOARD ENRICHMENT - MODERN HS TESTING

Osmotic fragility is classic and is what the slide teaches. In modern clinical practice, **eosin-5-maleimide (EMA) binding by flow cytometry** is commonly used because it is more standardized/sensitive in many centers. For this course, know the slide's osmotic-fragility principle and recognize EMA as enrichment.

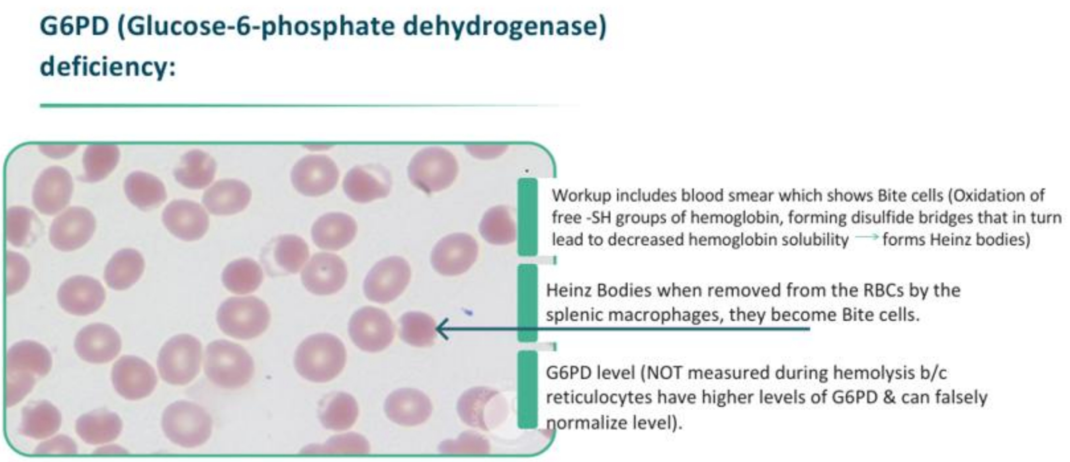
G6PD deficiency



Lecture slide 29 - full slide retained. G6PD feeds the pentose phosphate pathway to generate NADPH, which maintains reduced glutathione and protects hemoglobin/membranes from oxidative injury.

MECHANISM IN ONE CHAIN

Low G6PD -> low NADPH -> impaired reduced glutathione -> oxidative Hb damage -> Heinz bodies -> splenic removal of precipitates -> bite cells + episodic hemolysis.



Lecture slide 31 - full slide retained. The slide combines morphology with a major testing trap: measuring G6PD during an acute episode can be falsely normal because older deficient RBCs have lysed and the remaining reticulocytes contain more enzyme.

G6PD TRIGGERS

Infection, oxidant medications, fava beans.

The lecture specifically flags **rasburicase** as a potentially severe trigger and recommends checking G6PD status before use.

Slide 32: treatment

- Stop/avoid the oxidant trigger.
- Supportive care and hydration.
- Transfuse if the hemolysis is severe enough.

LO 5 checkpoint

Can you distinguish HS from G6PD by mechanism, smear, DAT, trigger pattern, testing caveat, and treatment strategy?

Slides 33-39

Autoimmune hemolytic anemia: warm vs cold

LO 6

Slide 34: both are immune hemolysis, but the effector mechanism differs

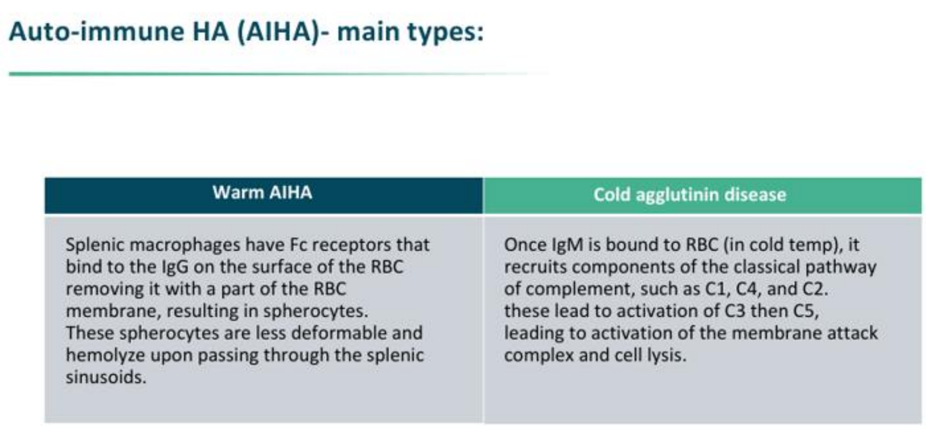
- Both can present with anemia, jaundice, and splenomegaly.
- Cold agglutinin disease can produce **acrocyanosis** in cooler extremities.

Auto-immune hA (AIHA)- main types:

Warm AIHA	Cold agglutinin disease
Due to auto-antibodies present and active at normal body temperature.	Due to auto antibodies active at temperatures below normal core body temperature
IgG antibodies	IgM antibodies
Ab are polyclonal pan agglutinins directed against common RBC antigens.	Directed against "I" or "i" antigens of the I blood group
Associated with infections (HIV, EBV, HCV), lymphoproliferative disorders (e.g. CLL), autoimmune disorders (e.g. SLE)	Can be induced by infections like mycoplasma
May also be drug induced e.g. with penicillins (haptens)	Must do workup to exclude lymphoma as an underlying cause

Lecture slide 35 - full slide retained. Warm AIHA is driven primarily by IgG active at body temperature; cold agglutinin disease uses IgM active at lower temperatures and strongly recruits complement.

Feature	Warm AIHA	Cold agglutinin disease
Antibody	IgG	IgM
Temperature	Body temperature	Cooler temperatures
Main mechanism	Fc-mediated splenic removal -> spherocytes	Complement recruitment; C3 deposition +/- intravascular lysis
Associations in deck	SLE, CLL/lymphoproliferative disease, infections, drugs	Mycoplasma and underlying lymphoid disease workup
DAT pattern	IgG +/- C3d	Usually C3d
Treatment emphasis	Steroid-based immunosuppression; rituximab	Avoid cold; complement-targeting therapy; treat cause



Auto-immune HA (AIHA)- main types:

Warm AIHA	Cold agglutinin disease
Splenic macrophages have Fc receptors that bind to the IgG on the surface of the RBC removing it with a part of the RBC membrane, resulting in spherocytes. These spherocytes are less deformable and hemolyze upon passing through the splenic sinusoids.	Once IgM is bound to RBC (in cold temp), it recruits components of the classical pathway of complement, such as C1, C4, and C2. these lead to activation of C3 then C5, leading to activation of the membrane attack complex and cell lysis.

Lecture slide 36 - full slide retained. Warm IgG-coated cells are partially phagocytosed by splenic macrophages, producing spherocytes. Cold IgM recruits classical complement components and can progress to membrane-attack-complex injury.

SPHERECYTE TRAP

Auto-immune HA (AIHA)- treatment:

Warm AIHA	Cold agglutinin disease
Antibody mediated cytotoxicity	• Complement mediated cytotoxicity
Rx aims at ↓ antibodies production: • Prednisone 1mg/kg, continue till Hgb is >10 g/dL and hemolysis labs are normal then taper over another 2-3 weeks. • Rituximab (anti-CD20 antibody)	Rx aims at inhibiting the activated complement system: • Avoid cold • Anti-C1 monoclonal antibody sutimlimab • Plasmapheresis or IVIG as a temporizing measure till anti-complement is started
• Treat underlying cause if identified • Folic acid supplementation	

Lecture slide 39 - full slide retained. The deck emphasizes prednisone/rituximab for warm disease and cold avoidance plus the anti-C1 antibody sutimlimab for cold agglutinin disease, with underlying-cause treatment and folate support.

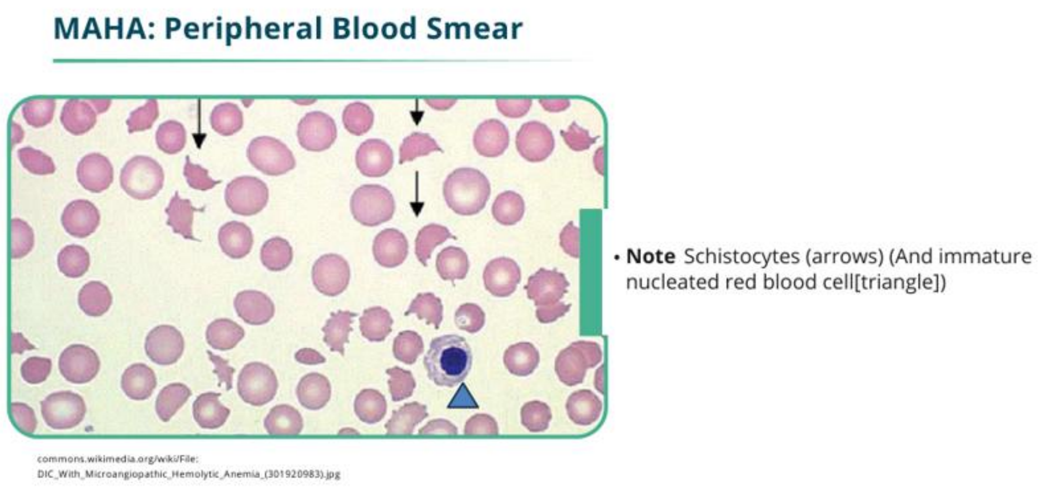
LO 6 checkpoint

Can you explain why warm AIHA makes spherocytes, why cold disease is complement-heavy, what each DAT looks like, and why their treatments differ?

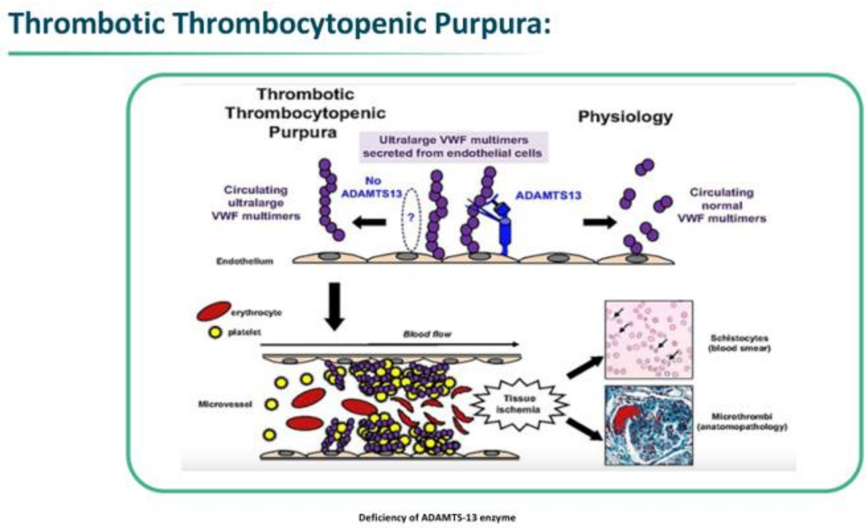
Slides 40-50 **MAHA and TTP: the emergency block** LO 4 + LO 6

Slides 40-43: mechanical fragmentation + thrombocytopenia is a red flag

- MAHA forms when RBCs are mechanically sheared in abnormal small-vessel environments.
- The lecture lists **TTP, atypical HUS, intravascular devices, and even severe B12 deficiency** among MAHA-like causes.
- Schistocytes are the morphology clue.



Lecture slide 43 - full slide retained. Fragmented helmet-like RBCs are the visual emergency clue. In a patient with thrombocytopenia and neurologic/renal abnormalities, this should trigger an urgent TMA/TTP workup.



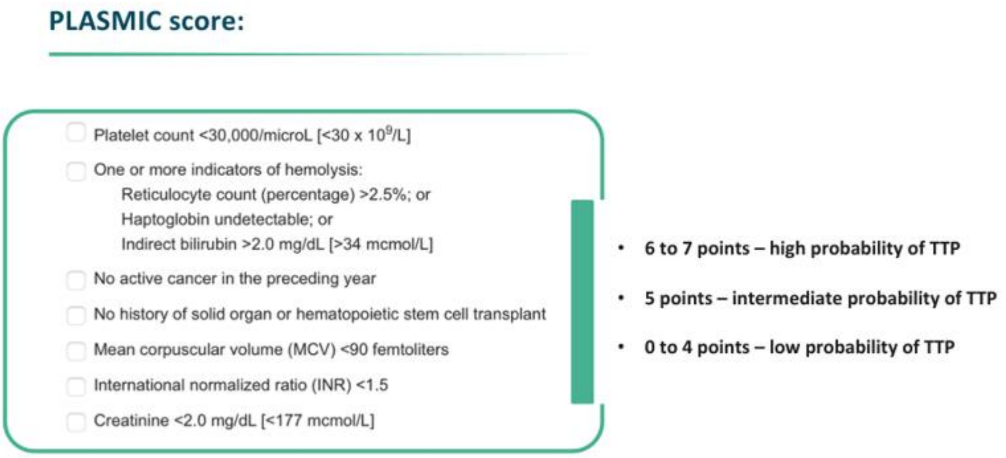
Lecture slide 44 - full slide retained. Without adequate ADAMTS13 activity, ultralarge vWF multimers persist, bind platelets, and form platelet-rich microthrombi that shear RBCs and cause tissue ischemia.

TTP MECHANISM

Severe ADAMTS13 deficiency -> ultralarge vWF multimers persist -> platelet microthrombi -> thrombocytopenia + MAHA + ischemia.

MODERN CLINICAL TRAP - DO NOT WAIT FOR THE PENTAD

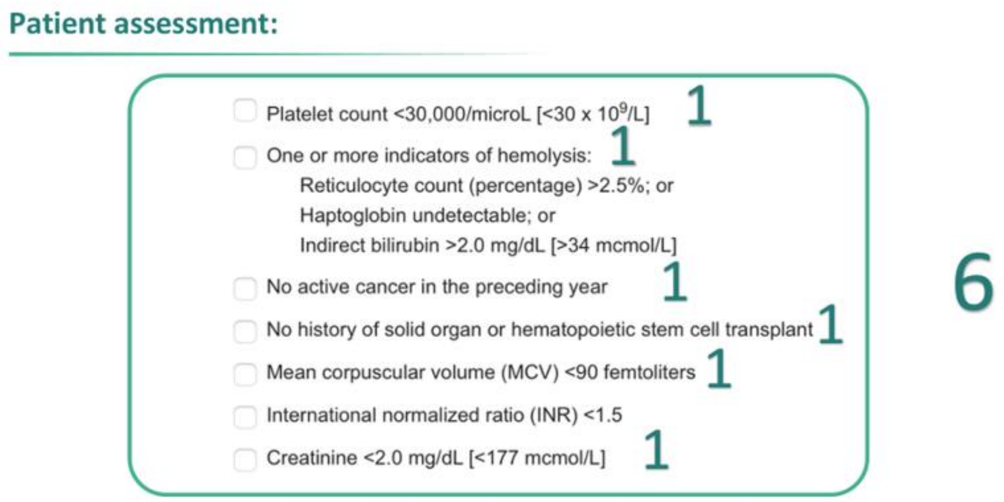
The classic TTP pentad is useful for memory, but patients do **not** need all five findings before treatment. The lecture's practical approach uses thrombocytopenia + MAHA and the PLASMIC score to estimate the likelihood of severe ADAMTS13 deficiency.



Lecture slide 46 - full slide retained. The slide uses seven features to estimate TTP probability: 6-7 high, 5 intermediate, 0-4 low. Its strength is sensitivity - useful for rapidly identifying patients who should be treated while ADAMTS13 is pending.

Case 2: Dalgetty walks the score

A 25-year-old woman with fever/confusion, Hb 7.8, platelets 23K, retic 4.1%, undetectable haptoglobin, normal-ish creatinine, and 3+ schistocytes is scored as high probability for TTP.



Lecture slide 49 - full slide retained. The faculty case receives 6 points, placing the patient in the high-probability category and justifying immediate empiric therapy while ADAMTS13 testing is sent.

FACULTY EMERGENCY MANAGEMENT - SLIDE 50

Do not wait for ADAMTS13 to return. The lecture says to start empiric **plasma exchange**, plus steroids and rituximab, with caplacizumab considered/added in the treatment strategy.

EMERGENCY VIGNETTE

Schistocytes + thrombocytopenia + neurologic findings + normal-ish coagulation studies -> think TTP -> plasma exchange now.

BOARD DISTINCTION - TTP VS DIC

TTP is a platelet/vWF microthrombus disorder, so **PT and aPTT are often normal**. DIC consumes coagulation factors, so coagulation tests are often prolonged with low fibrinogen and high D-dimer. This distinction is emphasized in the companion practice questions and is extremely testable.

LO 4 checkpoint

Can you identify TTP before ADAMTS13 returns, explain why schistocytes form, and state the immediate treatment without waiting for confirmatory testing?

Official LO 7 gap-fill Hemolytic transfusion reactions LO 7

WHY THIS SECTION IS HERE

The official learning objectives and the slide 2 objectives require distinguishing hemolytic transfusion reactions, but the 51-slide deck does not develop them before ending with TTP and questions. The uploaded companion handout supplies the acute-vs-delayed framework below.

Feature	Acute hemolytic transfusion reaction	Delayed hemolytic transfusion reaction
Timing	During / very soon after transfusion	Usually days later; companion gives about 3-10 days
Typical mechanism	ABO incompatibility, preformed antibody, complement-mediated **intravascular** hemolysis	Anamnestic IgG response to a previously encountered minor RBC antigen; usually **extravascular**
Clinical clues	Fever, hypotension, flank/back pain, hemoglobinuria, acute Hb fall	Unexpected Hb decline, mild jaundice, sometimes relatively subtle symptoms
DAT	Often positive	Often positive as newly detectable antibody coats transfused RBCs

NEED TO MEMORIZE

Acute = ABO / complement / intravascular / sick during transfusion.
Delayed = memory response to minor antigen / IgG / extravascular / Hb falls days later.

CLINICAL FIRST MOVE - ENRICHMENT

If an acute transfusion reaction is suspected, **stop the transfusion immediately**, keep IV access with appropriate fluid, verify patient/unit identity, and notify the blood bank while evaluating for hemolysis. This is standard clinical safety logic; it is not developed in the slide deck itself.

LO 3-4 synthesis — The hemolysis algorithm to run on a vignette — LO 3 + LO 4

#	Question	What it means
1	**Is there true hemolysis?**	Anemia + retic up + LDH up + indirect bilirubin up + haptoglobin down.
2	**Is this an emergency?**	Low platelets/schistocytes, renal failure, thrombosis/neuro findings, severe anemia, recent transfusion -> urgent TMA/TTP or transfusion-reaction pathway.
3	**What does the smear show?**	Spherocyte, schistocyte, bite cell, organism, ghost cell, etc.
4	**Intravascular or extravascular?**	Hemoglobinuria/profound haptoglobin fall favors intravascular; splenomegaly/spherocytes favors extravascular.
5	**DAT positive?**	Yes -> immune hemolysis. No -> membrane/enzyme/mechanical/infectious/other nonimmune causes.
6	**What exposure/history closes the diagnosis?**	Family history, oxidant drug/fava beans, cold exposure/Mycoplasma, autoimmune/CLL, prosthetic valve, transfusion, etc.

Spherocytes

DAT negative + family history/high MCHC -> HS.
DAT positive -> warm AIHA.

Bite cells

Oxidant trigger -> G6PD deficiency. Do not trust a normal G6PD assay during the acute attack.

Schistocytes

Check platelets + PT/aPTT + renal/neuro state. TTP is an emergency.

Dark urine after transfusion

Think acute intravascular hemolytic transfusion reaction until proven otherwise.

Learning-Objective Synthesis

Official LO	What you should be able to do without notes
LO 1	Define accelerated RBC destruction and derive retic/LDH/indirect bilirubin/haptoglobin changes.
LO 2	Classify hemolysis by location, tempo, heredity, intrinsic/extrinsic mechanism, and immune status.
LO 3	Use history, smear, emergency features, DAT, and targeted testing to identify the mechanism.
LO 4	Recognize TTP/MAHA and acute transfusion reaction as time-critical diagnoses.
LO 5	Explain HS and G6PD mechanisms, diagnostic clues, testing caveats, and treatment.
LO 6	Contrast warm AIHA and cold agglutinin disease mechanisms, DAT patterns, associations, and therapy.
LO 7	Distinguish acute ABO/complement-mediated intravascular reaction from delayed anamnestic IgG/extravascular reaction.

One-table comparison

Disorder	Smear / DAT	Mechanism	Key treatment logic
HS	Spherocytes; DAT negative	Membrane-cytoskeletal loss -> splenic trapping	Support/folate; splenectomy in selected disease
G6PD	Bite cells; DAT negative	Low NADPH/reduced glutathione -> oxidant damage	Avoid trigger, hydrate, transfuse if severe
Warm AIHA	Spherocytes; DAT IgG +/- C3	IgG-mediated splenic destruction	Steroid-based therapy; rituximab
Cold agglutinin	Agglutination; DAT C3d	IgM -> complement	Avoid cold; complement-directed treatment; address cause
TTP	Schistocytes; DAT negative	Severe ADAMTS13 deficiency -> platelet-vWF microthrombi	**Immediate plasma exchange** + immunosuppression/ caplacizumab strategy
Acute transfusion reaction	Hemolysis; DAT often positive	Usually ABO/ complement intravascular lysis	**Stop transfusion immediately** + blood bank evaluation/support

LAST 60 SECONDS

Hemolysis labs: retic up, LDH up, indirect bili up, haptoglobin down.
Spherocyte: HS vs warm AIHA -> DAT distinguishes.
Bite cell: G6PD; don't test during the attack if you can avoid it.
Cold: IgM + complement. **Warm:** IgG + spleen.
Schistocytes + low platelets: TMA emergency; TTP -> ADAMTS13 -> plasma exchange now.
Acute transfusion reaction: ABO/complement/intravascular. **Delayed:** anamnestic IgG/extravascular.

Source hierarchy: Dalgetty's 51-slide Hemolytic Anemias deck is the primary slide-order source. The official Week 7 objectives define emphasis. The uploaded Lecture 5 companion is used as supplemental explanation, particularly for transfusion reactions, because the slide deck names that objective but does not develop it. Board/clinical enrichment is labeled rather than presented as slide content.